

THE ALGEBRAIC GROUP MODEL is a security model for reasoning about adversaries that manipulate group elements algebraically.

i.e., they combine known elements via group operations and are forced to reveal how each new element is derived.

Definition: An algorithm A is called algebraic if:

- INPUT: Given a list $L = (L_0, L_1, \dots, L_t)$ of group elements from a cyclic group G of prime order q with generator g ,

- OUTPUT CONSTRAINT: Whenever A outputs a new group element $Z \in G$, it must also output a representation vector $z = (z_0, z_1, \dots, z_t) \in \mathbb{Z}_q^{t+1}$ such that:

$$Z = \prod_{i=0}^t L_i^{z_i} = L_0^{z_0} \cdot L_1^{z_1} \cdot \dots \cdot L_t^{z_t}$$

Meaning, the adversary must explain how every output group element is constructed from known elements using only group operations.

EXAMPLE: Consider a cyclic group $G = \langle g \rangle$ of prime order q . An algebraic adversary receives:

- $L_0 = g$ (generator)
- $L_1 = g^x$ (for unknown x)
- $L_2 = g^y$ (for unknown y)

If the adversary outputs $Z = g^{xy}$, it must provide (z_0, z_1, z_2) such that:

$$g^{xy} = g^{z_0} (g^x)^{z_1} (g^y)^{z_2} = g^{z_0 + z_1 x + z_2 y}$$

This forces: $xy = z_0 + z_1 x + z_2 y \pmod{q}$

To output g^{xy} , we must also output z_0, z_1, z_2

showing how g^{xy} equals $g^{z_0} (g^x)^{z_1} (g^y)^{z_2}$.

If the adversary doesn't know x or y , they can't find such z_i satisfying the equation above, since xy cannot be broken into linear sum of x and y .

Eg:- Imagine trying to write 6 as a combination of 1, 2 and 3. i.e., find z_0, z_1, z_2 such that $6 = 1z_0 + 2z_1 + 3z_2$. It is possible! $6 = 0 \cdot 1 + 0 \cdot 2 + 2 \cdot 3$.

But now imagine 6 is really the product of two unknown numbers x and y and now we know only 1, x and y . We want to now represent the product

xy as a linear combination $z_0 + xz_1 + yz_2$ without knowing x and y .

It is impossible to write it exactly as any linear sum of $1, x$ and y unless we know something about x and y .

In AGM, adversary must provide not just any new group element they output, but also a precise algebraic explanation (in exponents) of how that element was constructed from the unknown group elements to the challenger. If adversary cannot do so, then in practice it cannot produce that group element in the required way.

This allows the challenger to analyze, relate or even set up equations involving the exponents, which is the key to tight proofs.

